Venkata Lakshmi Parimala Pasupuleti

Software Engineer · PropTech · Python · Full-Stack · End-to-End Product Ownership · AI-Assisted Development · APIs · Cross-Functional Delivery · Real Estate Domain

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PROFESSIONAL SUMMARY

Software engineer with 8 years building tools that real estate and housing professionals rely on - the exact domain Zillow Metro owns. At Entrata, I shipped 20+ production modules used by property managers, leasing agents, and compliance teams across 100K+ housing units: income eligibility workflows, document management, compliance dashboards, and the async pipelines that keep them running at 99.5% uptime. I owned these end-to-end - from ambiguous business requirement through cross-functional scoping with product and operations, through production deployment and post-launch iteration - across 120+ Agile sprints. Two AI systems extend this pattern into product: AgentProbe (open-source LLM evaluation platform, CI-integrated regression testing, zero regressions across 1,240+ test cases) and LeaseI (8-agent agentic AI platform for lease compliance, built solo from concept through production using Claude Code as a daily accelerator). Full-stack across Python, TypeScript, JavaScript, React, and FastAPI. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Full-Stack & Product Engineering: Python (8yr), TypeScript, JavaScript, React.js, FastAPI, Flask, REST APIs, System Design, Database Design & Optimization, PostgreSQL, MySQL, Oracle, Redis, NoSQL, Docker, CI/CD (GitHub Actions), Testing Strategy (Unit, Integration, End-to-End), AWS (S3, EC2, RDS, Glue), RabbitMQ, Microservices, Linux, Git

AI & Development Tools: Claude Code, AI-Assisted Development, LangChain, LangGraph, Agentic Workflows, LLM Evaluation, LLM-as-Judge, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Anthropic API, OpenAI API, MCP, HITL, Guardrails, OCR, BERT, spaCy, NLP, PyTorch, scikit-learn

Delivery & Cross-Functional: End-to-End Product Ownership, Cross-Functional Collaboration (Product, Design, Data, Compliance, Operations), Operational Readiness, Production Monitoring & Observability, Performance Engineering, PHI Compliance, Data Governance, Code Review, Mentoring, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Scoped and shipped a production agentic AI system (LangGraph, FastAPI, Python) from ambiguous requirements to deployment - used Claude Code and AI-assisted development throughout to compress the build cycle; owned testing strategy, monitoring, and stakeholder handoff end-to-end.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through statistical validation across 190+ countries; designed observability dashboards improving research comprehension by 40%.
- Mentored 600+ students in Python, SQL, system design, and API development; conducted technical workshops and provided cross-functional support across multiple course sections.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **End-to-end product ownership in PropTech** - 20+ production modules that property managers, leasing agents, and compliance teams relied on daily; owned HUD income-limit pipeline 7 years (federal data → 180,000 domain records powering eligibility and rent calculations for 100K+ housing units); went where the problem lived - backend services, APIs, data models, compliance workflows, and the async processing layer - across 120+ Agile sprints with cross-functional partners in product, compliance, and operations.
- **Full-stack systems and APIs at 99.5% uptime** - Python and JavaScript/TypeScript backends; FastAPI REST APIs and integration layers; RabbitMQ event-driven processing with retry, idempotency, and dead-letter guarantees; Docker containerization (50% scalability improvement); AWS S3, EC2, RDS, Glue; Redis caching; 200+ SQL query optimizations (30% performance gain); 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents).
- **Testing strategy, code quality, and mentoring** - unit, integration, and regression testing with CI/CD via GitHub Actions; maintained sub-10 bug count for 6+ consecutive months; 300+ peer code reviews enforcing standards; leveraged Claude Code and AI-assisted development to accelerate debugging and iteration; mentored 5+ engineers on Python, system design, and production reliability; followed work into production and iterated on metrics and customer behavior.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for property management and affordable housing compliance; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Agentic AI Platform - Concept Through Production

PropTech · Full-Stack · End-to-End

Python · LangGraph (8-agent) · FastAPI · React · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface AI platform for affordable housing compliance - built solo from ambiguous problem through production using Claude Code as daily accelerator: document intelligence (OCR, section-aware chunking, citation verification), 8-agent LangGraph support investigation (tool routing, memory, HITL escalation), and tenant self-service with deterministic guardrails. Full testing strategy and measurable outcomes: 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

AgentProbe: LLM Evaluation with CI-Integrated Testing

Evals · Testing · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform shipped end-to-end: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, multi-provider routing. Post-launch iteration on test coverage - zero regressions shipped across 1,240+ test cases. Open-source.

AI on FHIR: Full-Stack NLP Query System

Full-Stack · APIs · React

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack application (Python backend + React.js/TypeScript frontend) translating natural-language queries into FHIR REST API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017